Data lakes and data warehouses

A data lake is a repository that stores all your organisation's data (structured, semi-structured and unstructured) while data warehouses are locations where different data sources may be combined to support specific business intelligence and analytics needs.

Data architecture documents an organisation's data assets, maps how data flows through its systems, and provides a blueprint for managing data.

How is data management different from data governance?

Data governance is a set of procedures and guidelines on data access, use and management. Data management includes processes and tools to ingest, store, catalogue, prep, explore and transform data, so that you can use it for decision-making. Also, data governance is a practice to increase the business value of data without compromising its security, integrity, or privacy while data management is a process that supports data consumption by following those data governance guidelines.

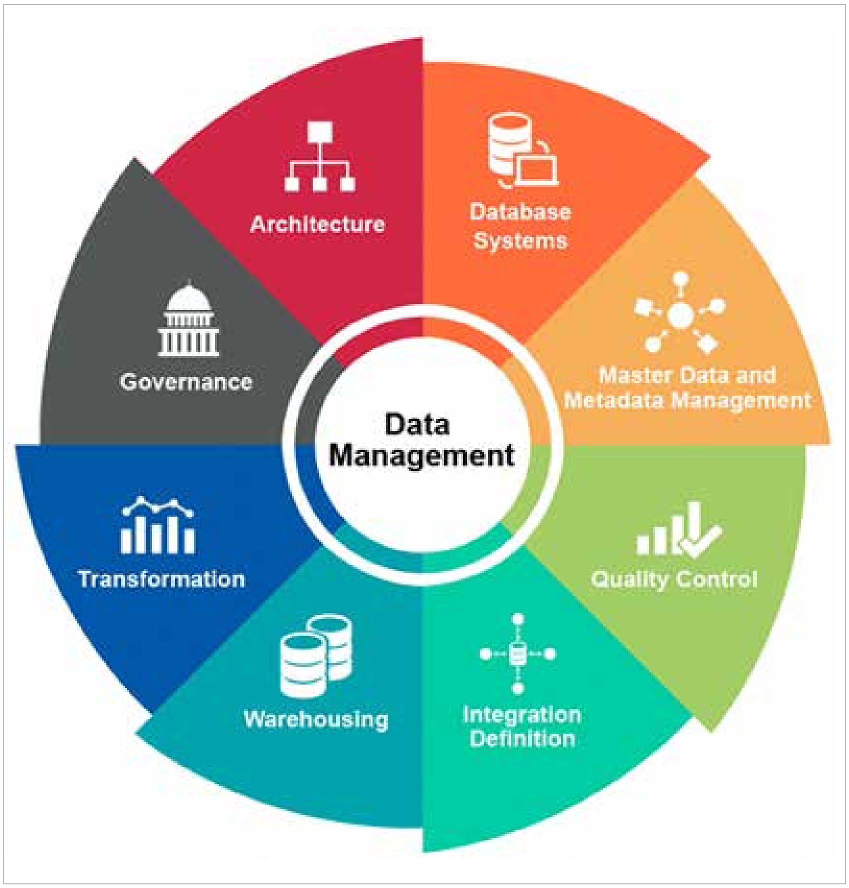

Figure 1: Data management

Data management tools

ETL and data integration tools

ETL is the process of copying data from multiple sources into a destination system that represents the data. Data integration, on the other hand, refers to the process of combining data from multiple sources into a single destination.

Hevo Data: This tool allows you to replicate data in near real-time from multiple sources (including Snowflake, BigQuery, Redshift and Databricks) to the destination.

Talend Open Studio: Talend is one of the best data management software that provides corporations with data integration, integrity and preparation. It is an open source data integration solution that was launched by Talend in October 2006. Data transformation features offered by Talend include filtering, flattening/normalising, aggregating, replicating, joining and temporal windowing.

Cloud data management tools

There are a lot more data warehousing and management options available now that storage and bandwidth are less expensive. Cloud-based technologies have been embraced by companies that need to store, analyse and sort through massive amounts of data in order to be more efficient. The development of strong cloud data management solutions over the last five to ten years has made this feasible. Although these industries are dominated by behemoths like Google and Amazon, technology has allowed numerous smaller businesses to provide tools for clients with varying data demands. Three well-known enterprise data management tools are as follows.

Amazon Web Services: Multiple tools from AWS can be integrated to create a cloud data management track. This Amazon subsidiary offers pay-as-you-go cloud computing platforms and APIs on demand. The following are a few services provided by Amazon Web Services:

- Amazon Redshift for data warehousing
- Amazon Athena for SQL-based analytics
- Amazon Quicksight for dashboard and data visualisation
- Amazon Glacier for long-term backup and storage
- Amazon S3 (Simple Storage Service) for temporary and intermediate storage